

SANNICOLAU MARE, Romania

WMO: 151990

Lat: 46.0714N

Lon: 20.6017E

Elev: 283

StdP: 14.55

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.6	17.8	9.4	9.0	16.2	14.0	11.3	21.4	25.3	27.4	22.2	35.7	4.6	0	0.487

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	22.5	93.3	70.5	90.2	70.5	87.2	69.9	75.3	83.7	73.4	83.5	71.8	83.1	7.2	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.3	125.1	78.8	70.7	114.3	76.7	68.7	106.4	75.1	39.0	83.1	37.2	83.6	35.7	83.0	82.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.6	18.9	16.1	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	53.6	31.3	37.4	43.7	53.5	63.0	69.6	71.9	73.3	64.3	54.2	45.0	35.1
	DBStd	16.10	8.05	8.40	7.89	8.42	6.44	6.00	5.69	6.10	7.54	7.85	7.82	7.57
	HDD50	1921	580	355	221	62	2	0	0	0	48	190	463	
	HDD65	4954	1045	772	661	351	113	23	10	7	105	340	600	927
	CDD50	3236	0	2	25	166	404	589	678	721	431	179	40	1
	CDD65	793	0	0	0	5	50	163	222	262	85	6	0	0
	CDH74	8454	0	0	2	104	544	1641	2438	2847	795	83	0	0
	CDH80	3271	0	0	0	9	110	573	1001	1312	263	3	0	0
(14) Wind	WSAvg	6.7	6.4	7.2	7.7	8.1	6.9	5.9	6.1	5.8	6.1	6.0	6.9	6.8
(15) Precipitation	PrecAvg	21.8	1.3	1.1	1.1	1.6	2.3	2.8	2.5	1.7	2.0	1.9	1.7	1.8
	PrecMax	32.0	2.3	3.0	4.4	4.1	6.1	8.1	5.9	3.5	5.1	4.5	3.9	3.6
	PrecMin	10.4	0.2	0.0	0.0	0.2	0.4	0.5	0.7	0.2	0.2	0.1	0.1	0.1
	PrecStd	5.4	0.7	0.7	0.9	0.9	1.4	1.8	1.2	1.0	1.1	1.1	0.9	1.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.2	60.6	71.7	81.0	86.4	92.8	96.0	98.3	91.7	79.4	68.2	55.7
		MCWB	49.6	51.9	56.6	63.0	69.4	73.8	71.9	70.4	68.5	62.3	58.5	50.9
	2%	DB	47.4	56.6	65.4	77.1	83.2	89.6	92.4	94.3	86.6	76.2	63.1	50.8
		MCWB	44.3	50.0	53.7	60.6	67.1	72.3	71.3	69.2	67.5	60.7	56.2	48.0
	5%	DB	44.5	52.5	60.3	72.7	80.1	86.3	88.9	90.8	82.4	72.5	59.9	47.1
		MCWB	42.9	47.4	50.4	58.1	65.5	71.6	70.3	69.2	65.2	59.3	55.2	45.1
	10%	DB	41.9	49.1	56.2	68.8	76.8	82.9	85.5	87.4	78.7	67.7	56.6	44.3
		MCWB	40.6	45.7	47.5	56.0	63.5	69.8	68.7	69.1	64.8	57.6	52.4	42.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.8	54.3	58.9	65.0	74.7	79.3	76.3	75.0	73.4	64.6	59.9	52.9
		MCDB	53.3	59.1	70.0	77.3	81.3	87.0	82.4	84.3	86.4	73.7	63.9	55.1
	2%	WB	45.3	51.2	55.0	62.4	69.5	76.4	74.4	73.2	69.3	62.5	57.7	48.7
		MCDB	46.9	54.9	64.2	75.1	78.4	84.0	83.8	85.6	81.6	72.0	62.2	51.1
	5%	WB	43.1	48.4	52.0	60.3	67.2	73.8	72.7	71.8	67.4	60.7	55.6	45.8
		MCDB	44.5	52.3	59.0	70.5	76.1	81.9	84.4	84.0	77.8	70.1	60.3	47.5
	10%	WB	40.8	45.8	49.3	58.0	65.2	71.7	71.0	70.5	65.6	58.8	53.1	43.4
		MCDB	42.2	48.6	55.2	67.2	73.5	80.3	83.5	82.8	74.4	66.5	56.3	44.7
(35) Mean Daily Temperature Range	5% DB	MDBR	10.5	13.9	19.0	21.5	20.5	21.4	22.3	22.5	20.1	19.6	13.9	9.7
		MCDBR	13.4	20.1	25.6	27.9	25.3	24.6	26.1	27.1	25.6	26.4	18.2	14.2
		MCWBR	10.1	13.7	15.1	13.8	11.3	10.3	9.8	8.1	9.6	12.2	11.9	11.3
	5% WB	MCDBR	12.6	18.0	23.3	24.2	21.8	21.9	23.1	22.9	21.5	23.6	16.6	14.0
		MCWBR	10.3	12.9	14.8	12.5	11.2	11.1	10.7	9.1	9.0	11.7	11.3	11.3
(40) Clear-Sky Solar Irradiance	taub		0.341	0.366	0.401	0.456	0.440	0.436	0.448	0.452	0.430	0.393	0.362	0.340
	taud		2.342	2.281	2.215	2.065	2.182	2.225	2.189	2.183	2.213	2.310	2.349	2.357
	Ebn at Noon		235	252	259	254	264	265	259	252	246	239	224	221
	Edh at Noon		24	31	39	49	45	43	44	43	38	30	24	22
(44) All-Sky Solar Radiation	RadAvg		361	630	1032	1470	1783	1956	1940	1740	1214	818	477	284
	RadStd		40	92	136	140	163	154	149	162	133	86	40	42

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (56 neighbors)	+1.22	N/A	N/A	N/A	N/A	+0.70	N/A	-202	-310	+191	+103

Nomenclature: See separate page